

Post Hoc Analyses

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Abstract

In this paper we will be evaluating various post hoc tests through the lens of their strengths and weaknesses, especially through certain applications like drug testing.

1 Introduction

Comparing shared attributes of various objects, abstract or tangible, is so commonplace to everyday life that I have no doubts people give less thought than they should to the final decision, whether it be trying to find which is most effective, available, or some other desirable trait. Post hoc tests, and ANOVA testing, can remove the need for subjective bias in an important comparison, and give you concrete evidence that one is truly better than the rest. How though, can we find the post hoc test that performs best every time we need to make a unique comparison?

2 ANOVA Testing

The ANOVA test is a necessary prelude to post hoc testing. ANOVA tests will examine our various items and determine if there is simply enough evidence to claim there is a difference among all the items. Later, post hoc tests will tell us where those differences lie, and in what direction. How though, does the ANOVA test accomplish this?

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij},$$

- Y_{ij} : The observation for the j th individual in the i th group.
- μ : The overall (grand) mean of all observations across all groups.
- τ_i : The effect of group i , representing how group i 's mean differs from the overall mean.
- ε_{ij} : The random error term for the j th observation in group i , assumed to be independent and normally distributed with mean 0 and variance σ^2 , i.e. $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Getting a value for each observation in each group is the first part of the ANOVA test. In the next step, they will all be summed into different values and have their differences with other values compared so we can calculate our final F statistic.

$$SS_{\text{Total}} = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y})^2$$

$$SS_{\text{Treatment}} = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y})^2$$

$$SS_{\text{Error}} = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2$$

$$F = \frac{\text{Mean Square Treatment (MST)}}{\text{Mean Square Error (MSE)}} = \frac{SS_{\text{Treatment}}/(k-1)}{SS_{\text{Error}}/(N-k)}$$

where Y_{ij} is the observation j in group i , $\bar{Y}_i$ is the mean of group i , $\bar{Y}$ is the overall mean, n_i is the number of observations in group i , k is the number of groups, and N is the total number of observations.

Finally, we can calculate the p-value the ANOVA hypothesis test is concluded. Now that it is laid out, we'll examine our initial ANOVA test (only our first of 1000).

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_a : At least one μ_i differs from the others

F-Statistic:

$$F = 34.91582$$

P-value:

$$p = 3.252205\text{e-}08$$

P-value Interpretation: If there is truly no difference among the groups, we would expect data like ours, or more extreme, 3.252205e-06% of the time.

Conclusion: There is very strong evidence in favor of there being a difference among the groups.

Noticeably, this is very "extreme" evidence, enough so that we are magnitudes higher than what we would need to conclude there is a difference. This carries over to all the simulations of ANOVA that we ran, but what exactly did we run these ANOVA on?

3 Generating Data

Using the `rnorm` command in `posit`, we are able to generate 3 different data sets that we will use to evaluate our post hoc tests. We standardize the

sample size and standard deviation of all 3, only changing their means. This satisfies all our ANOVA assumptions, normality, independence, and equal variance of errors. The first two are satisfied logically, because `rnorm` generates a normal data set and each observation can't affect another, and homoskedasticity is satisfied because we have an equal standard deviation across all three groups.

One round of comparisons isn't enough to ensure random variance won't skew our data, however. We want to see, on average, how well each post hoc test can detect differences between our groups, and for that we will need more copies of each group. Using simulations, we generate 1000 copies of each group. That is, they'll have the same properties, same mean, sd, and sample size, but `rnorm` will randomly generate different values for each of the 10 observations, so we'll maintain our assumptions throughout.

4 Post Hoc Tests

Only after we've generated our data, can we actually run ANOVA on all 1000 groups of comparisons, but we've already discussed the process. With our ANOVA results, we'll walk through each of our 7 post hoc tests that we've chosen for this experiment.

One thing to take note of is that all the tests performed similarly, for the most part. Despite their many differences, all of the graphs look like they could be the same test, just changed by slight variations in the data. I

believe this is due to keeping a low standard deviation of 1 for all the data. It didn't allow for hardly any random variation that would actually make a big difference for a conservative test versus a liberal test. Still, because of how many repetitions we went through, it is still clear that our results tell us how well each performed.

4.1 Bonferroni Correction

A fitting first test, for it's simple calculation, and accompanying limited, elementary application. Useful for teaching statistics, but low power even by conservative test's standards.

	Prop Significant	Average P-value
Old vs New	0.394	2.631323e-01
Placebo vs New	1.000	6.733524e-05
Placebo vs Old	0.968	7.271139e-03

Table 1: Bonferroni Correction Summary Table

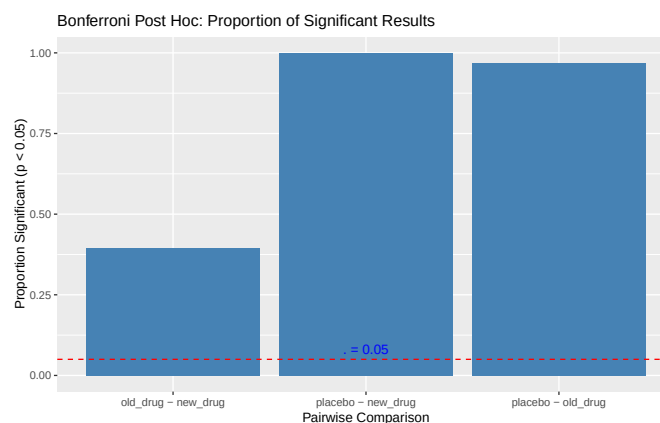


Figure 1: Bonferroni Correction Summary Graph

We see from the graph and the table that Bonferroni Correction Significantly underperforms compared to the later tests, correctly identifying less than 40% of the differences in Old vs New. While it is less pronounced, it also misses more differences in Placebo vs Old. Again, this is to be expected, as it is a simple and extremely conservative test.

4.2 Tukey's HSD

Another limited test, in the sense that it's best use would be for already simply testing, one mean vs. one mean, or more. Still conservative, but nearly anything is a step up from Bonferroni.

	Prop Significant	Average P-value
Old vs New	0.433	1.958669e-01
Placebo vs New	1.000	6.273578e-05
Placebo vs Old	0.974	5.958850e-03

Table 2: Tukey's HSD Summary Table

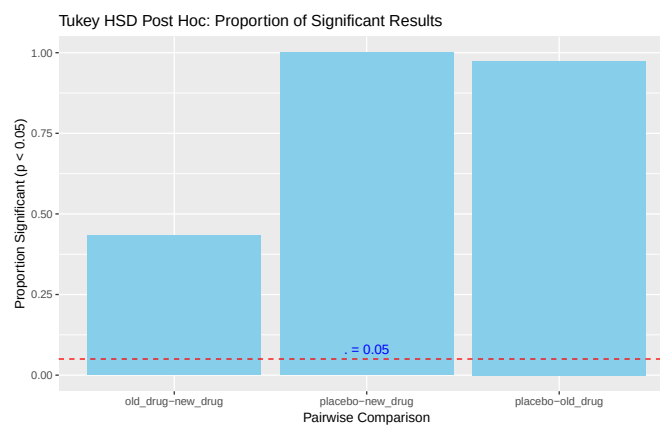


Figure 2: Tukey's HSD Summary Graph

Expectedly, Tukey's HSD performs better than the Bonferroni Correction, but still rather marginally. It's another conservative test, it makes sense for it to underperform when the only error we can generate is Type II.

4.3 Scheffe's Method

While Tukey's HSD can struggle with non-pairwise comparisons, Scheffe's method is generally preferred for it. It also has some traits that leave it preferred if assumption of equal variance is violated, but non-parametric tests would be better here. Conservative.

	Prop Significant	Average P-value
Old vs New	0.377	0.2112300948
Placebo vs New	1.000	0.0000931723
Placebo vs Old	0.960	0.0074852060

Table 3: Scheffe's Method Summary Table

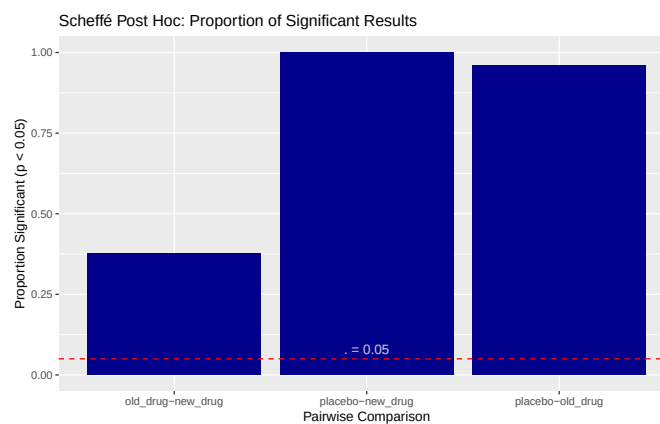


Figure 3: Scheffe's Method Summary Graph

Surprisingly, Scheffe's method manages to perform worse than Bonferroni,

but this can be explained. While Bonferroni Correction only protects the Family Wise Error Rate, or simply our three pairwise comparisons, Scheffe's method protects against all possible contrasts. Of course, this will inflate our Type II error rate, as we're essentially needlessly protecting against the risk of Type I errors, for comparisons we're not interested in.

4.4 Fisher's LSD

Our first liberal test, it offers much more power than the previous tests but also comes with a much higher chance of a type I error, or finding a difference when there isn't really one. In some cases, a type II error isn't so consequential, but in general it is safer to run the risk of a type II error more. It notably does not correct for multiple comparisons, so take heed with non-pairwise comparisons.

	Prop Significant	Average P-value
Old vs New	0.595	1.143169e-01
Placebo vs New	1.000	2.244508e-05
Placebo vs Old	0.991	2.423713e-03

Table 4: Fisher's LSD Summary Table

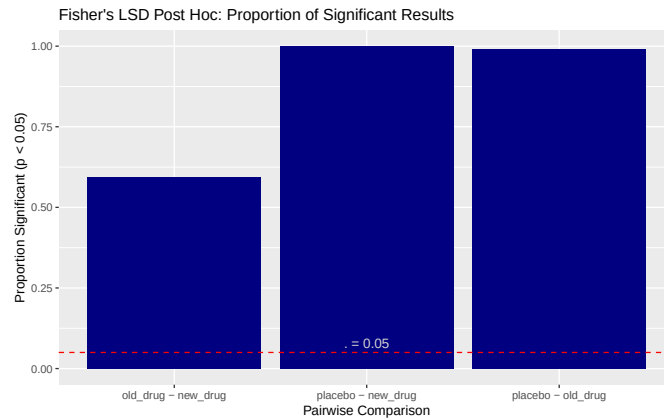


Figure 4: Fisher's LSD Summary Graph

As expected of a liberal test, it captures much more of the differences we know are there. For a test where we simulate data, and we know there are actual differences, a liberal test will always perform better. In real studies though, data is likely to not be simulated, so a balance of power is needed.

4.5 Holm-Bonferroni

Holm-Bonferroni is a conservative test, but with noticeably more power than Bonferroni Correction. Considered one of the best in terms of balancing power and preventing error. Simply, use this over Bonferroni Correction.

	Prop Significant	Average P-value
Old vs New	0.593	1.146882e-01
Placebo vs New	1.000	6.417125e-05
Placebo vs Old	0.990	3.264599e-03

Table 5: Holm-Bonferroni Summary Table

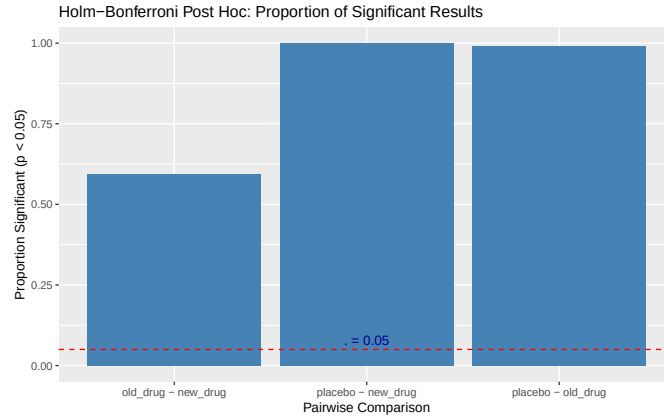


Figure 5: Holm-Bonferroni Summary Graph

In Holm-Bonferroni, we find one of the best tests for a balance of conservative estimation and power. Our pretext is backed up by our results, with it being one of our top performers. Due to its performance, and lack of complexity or specialization, it earns the spot of our chosen victor.

4.6 Dunnett's Test

A very specialized test, Dunnett's maintains considerable power while still being conservative, but is meant to only be applied when comparing placebo or control groups against an experiment group.

	Prop Significant	Average P-value
Old vs New	NA	NA
Placebo vs New	1.000	4.397824e-05
Placebo vs Old	0.985	4.516697e-03

Table 6: Dunnett's Test Summary Table

*Old vs New not ran to simulate the limitations of this test.

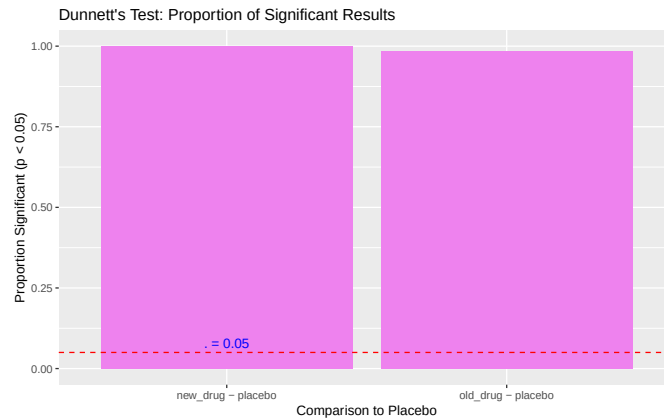


Figure 6: Dunnnett's Test Summary Graph

Dunnnett's Test very effectively captured the difference in Placebo vs Old group, but we notably did not run it for the Old vs New. This is because it is made to perform best when comparing an experiment group to a control group, so to not misapply it as well as show it's limitations, we only ran it for our two groups which involved the placebo.

4.7 Benjamini-Hochberg

Our final test, and definitely the most flexible, Benjamini-Hochberg has a customized formula that allows the performer to manually choose the false discovery rate, or how often false positives will occur during multiple tests. This lends it to being liberal in general, but you can always lower it to be more conservative. For our purposes here, we have set it to a 5% FDR.

	Prop Significant	Average P-value
Old vs New	0.594	1.144483e-01
Placebo vs New	1.000	6.233299e-05
Placebo vs Old	0.991	2.791599e-03

Table 7: Benjamini-Hochberg Summary Table

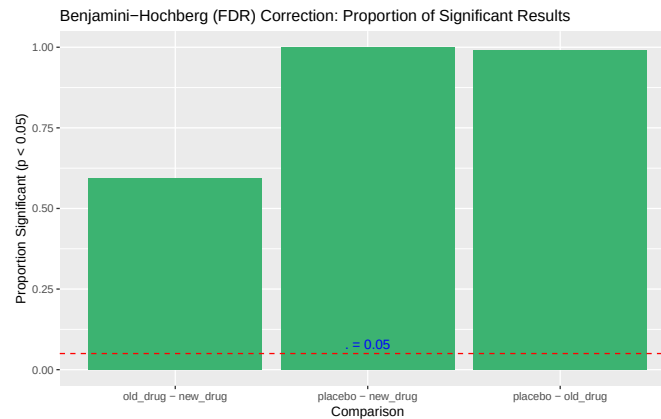


Figure 7: Benjamini-Hochberg Summary Graph

Benjamini-Hochberg is perhaps one of the most powerful tools covered in this examination, able to fit into nearly any experiment due to the ability to customize the FDR. However, this may open the door for misuse if, for example, one point higher of FDR would allow us to achieve statistically significant results.

5 Conclusion

Looking at the results of the later tests, we can see that many of them perform much better than the simpler tests. Even Fisher's LSD, a considerably liberal

test, only managed to correctly identify one more difference in the Placebo vs Old category compared to Holm-Bonferroni, a conservative test that is able to match that power. Benjamini-Hochberg also barely outperformed Holm-Bonferroni, but it's technically liberal nature may dissuade us from using it in experiments that concern commercial drugs, where Type I errors are more important to protect against.

References

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